PRIVACY-PRESERVING SHOPPING INTELLIGENCE

How much recommendation quality survives when shopping data stays on the device?

Cloud vs federated vs on-device learning for Microsoft Edge — progress update and decisions

Cloud Federated On-device

Ahmed Abdelhamed · MSc, Queen's University · Microsoft supervisor meeting · 30 September 2026

Status of all facts: as recorded 29 Sep 2026, 18:36 UTC

THE QUESTION

Three ways Edge could learn from shopping behaviour

“How much shopping recommendation quality can be retained when moving from cloud-based to on-device or federated models?”

CLOUD

Events go to the server

One large model is trained and served centrally.

Quality ceiling — but raw browsing leaves the device.

FEDERATED

Events stay on the device

The device trains locally and sends only model updates; the server averages them.

Optional differential privacy adds noise to the updates.

ON-DEVICE

Nothing is uploaded

A small model is shipped once and can be fine-tuned locally on the user's own history.

Strongest privacy; limited by one user's data.

Headline measure: retention = quality of a deployment ÷ quality of the cloud model, with a paired confidence interval.

Source: Research proposal §1; public study description (federated-deployment-study.md)

SCOPE

What changed from the proposal, and why

	Proposal	Now	Why
Dataset	Criteo, MovieLens, Amazon, Taobao, RetailRocket	REES46 e-commerce events (Oct–Nov 2019)	Public, per-user timestamped shopping sessions → each user can be simulated as one device
Task	Next category, purchase likelihood, best offers	Next product viewed, ranked over the full catalogue	One well-defined task, exact ranking, no sampled negatives
Metric	NDCG@10, HR@10, purchase AUC	MRR@20 per user (NDCG@10, HR@10 also reported)	Per-user ranking quality matches a browser serving one user
Models	Not fixed	GRU vs SASRec contest → SASRec	Backbone chosen by a pre-registered rule, not by preference
Federated	FedAvg, FedProx, FedBN, personalised FL	FedAvg, FedProx, personalised FL, 8-bit uploads; tuned server optimizers	FedBN not applicable: the model uses layer norm, not batch norm
Privacy	“Various privacy levels” (not defined)	Differential privacy at ε = 8 and ε = 1; a six-level privacy matrix	Makes “privacy level” measurable

Kept from the proposal: cloud upper bound, federated variants, on-device variants, cold-start analysis, persona analysis, communication cost, browser deployment check.

Source: Research proposal §§3–4; study description; project map; realism audit §F

DATA AND TASK

Next-product prediction over the whole catalogue

158,486

products ranked for every prediction

131,072

users in the main cohort (5–100 events each)

2.59 M

training decisions in the main cohort

1.35 M

users in the full pool (for the realistic cloud)

Chronological split — the model never sees the future

Training — 1 Oct to 15 Nov 2019	Validation — 16–22 Nov	Test — 23–29 Nov (sealed)

- Each user = one simulated device; a prediction uses at most the user's last 50 events.
- Calibration users (32,768, part of the main cohort) are the only place where tuning results may be read before the final analysis.
- The test week is opened once, after every method is locked. It contains Black Friday — disclosed and reported as a sub-period.

Source: Data protocol lock (via realism audit A1, project map §3); study description “Data and task”

METHOD

How we keep the comparison fair

1 Pre-registered

Every rule, selection criterion and contrast is written and hash-sealed before the result it governs is read. Changes are dated amendments.

2 Same budget, same everything

Same data version, features, evaluator and training exposure for every method; tuning happens only on calibration users.

3 Full-catalogue ranking

Exact ranking over all 158,486 products with a pessimistic tie rule; unknown items score zero.

4 Paired bootstrap

1,000 user-level resamples on the same users for every comparison; two seeds for the main methods.

5 One sealed test look

Validation is read once for all methods together, then everything is locked, then the test week is opened once.

6 Deterministic runs

Strict FP32, fixed seeds, frozen evaluator; the evaluator hash is part of every comparison key.

Source: Evaluation contract; study description “How fairness is kept”; multi-expert review strengths

DISCUSSION / DECISION 1

Is this the right yardstick?

1

Should retention be measured against the matched cloud or the full-pool cloud?

Matched cloud = same 131k users (isolates the learning method). Full-pool cloud = 1.35M users (what Edge would really run). We plan to report both — which should lead?

2

Is per-user MRR@20 the right headline metric for a browser feature?

NDCG@10 and HR@10 are reported too. Would a product team prefer a top-10 metric?

3

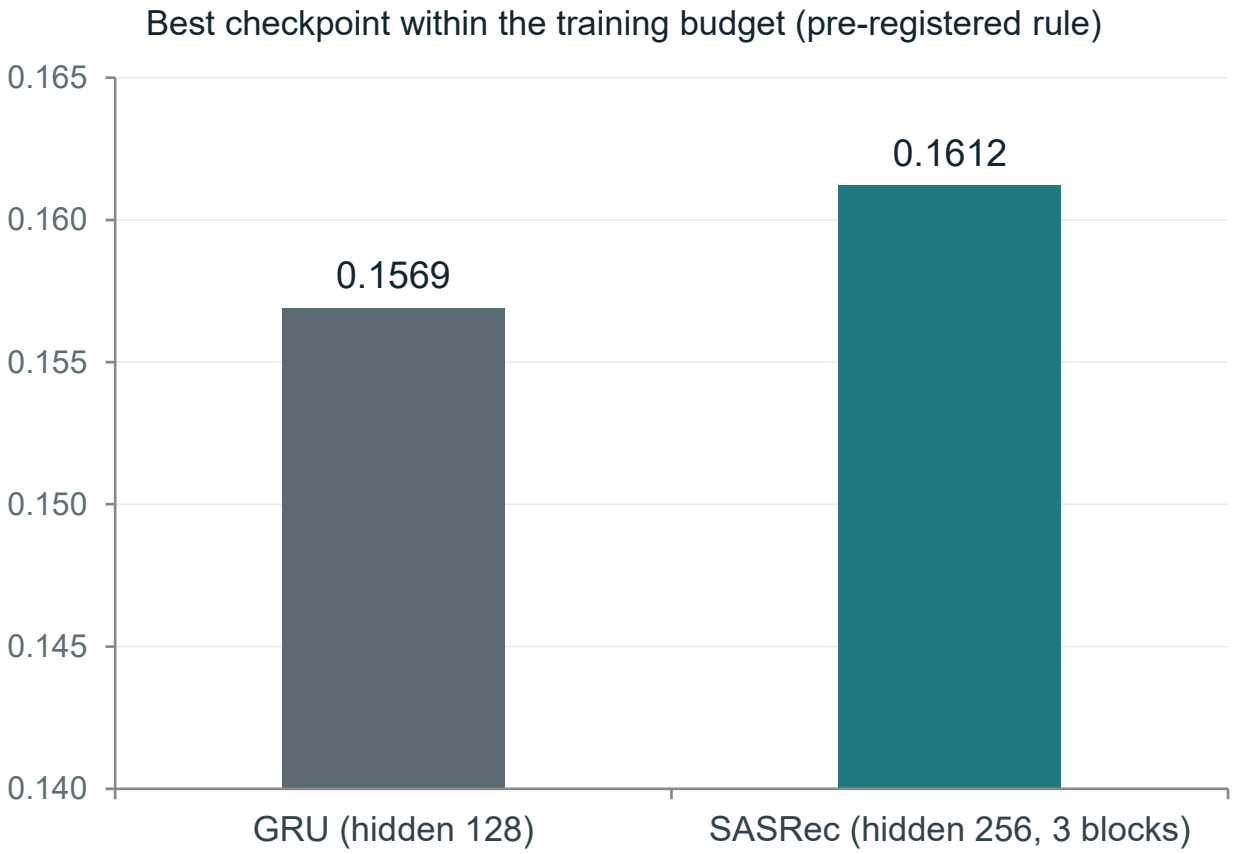
Are two seeds and one sealed test look enough for the headline methods?

A third seed costs about half a day of GPU time per method (estimate).

Context: the matched cloud saw 131,072 users; the full-pool cloud 1,348,682 users

BACKBONE CONTEST

SASRec won the pre-registered contest against GRU



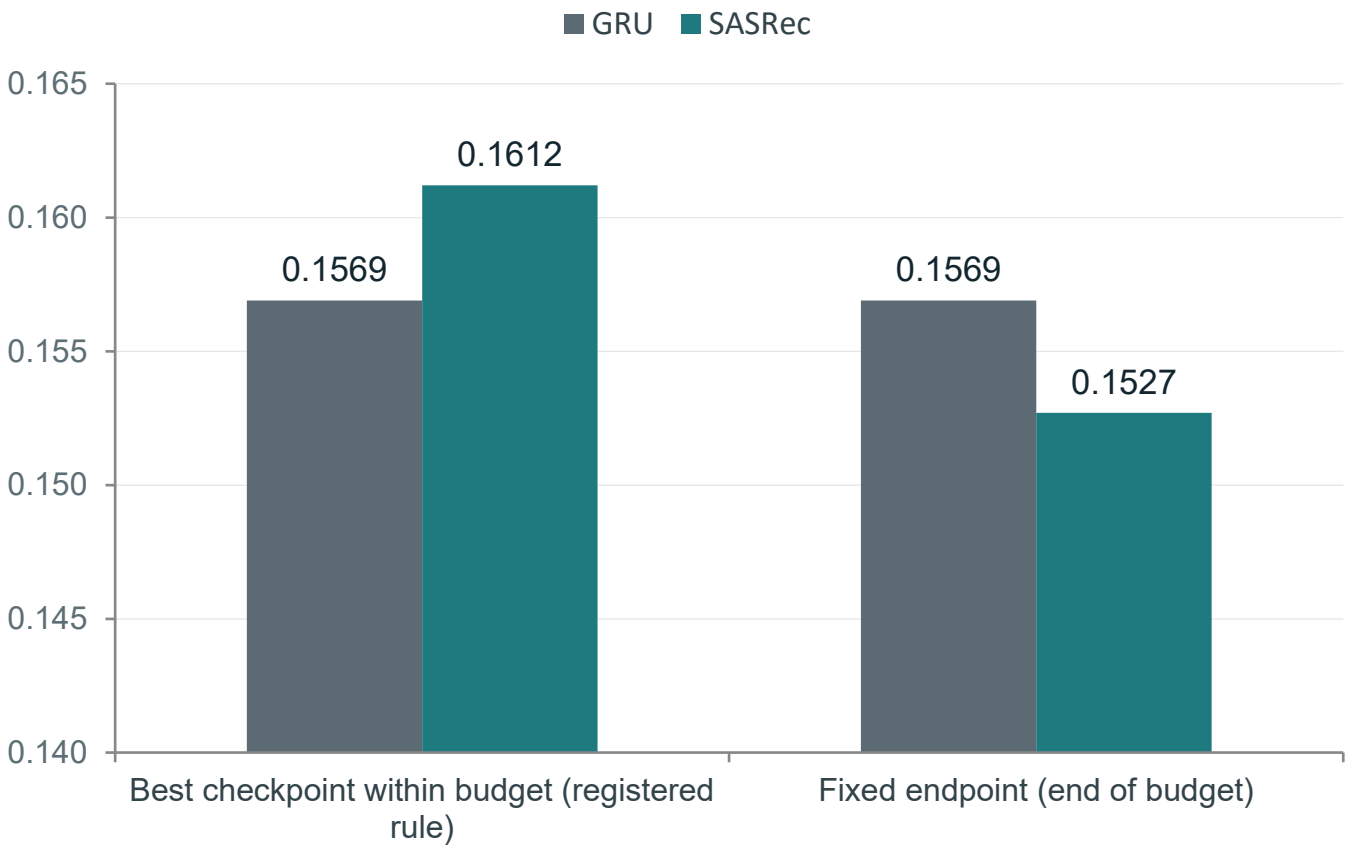
−0.0043

GRU − SASRec, 95% CI [−0.0052, −0.0033] — excludes zero; stable across both seeds

- 8 runs on the main cohort: 2 sizes per family × 2 seeds, equal tuning budget.
- NDCG@10 agrees: 0.1967 vs 0.1910.
- A check on users outside the calibration set agrees (no flip).
- Validation: 40,764 users, 519,714 predictions.

Source: Contest result file; decision log (contest run, 28 Sep)

The order reverses at the fixed training endpoint



What it means

- SASRec peaks halfway through the budget, then declines; GRU was still improving at the end.
- At the endpoint GRU is ahead: +0.0042, CI [+0.0033, +0.0052].
- Both rules were fixed before any result, so the verdict stands — but it is reported as “selected under the pre-registered rule”, never “SASRec beats GRU”.
- Not size-matched: 83.9M vs 21.0M parameters. A size-matched comparison is registered as descriptive.

Source: Contest result file; contest diagnostics (decision log, 28 Sep 14:05Z); contest review ruling

MODEL SIZES

A large cloud model, and a device model sized for a browser

CLOUD MODEL

83.9 M

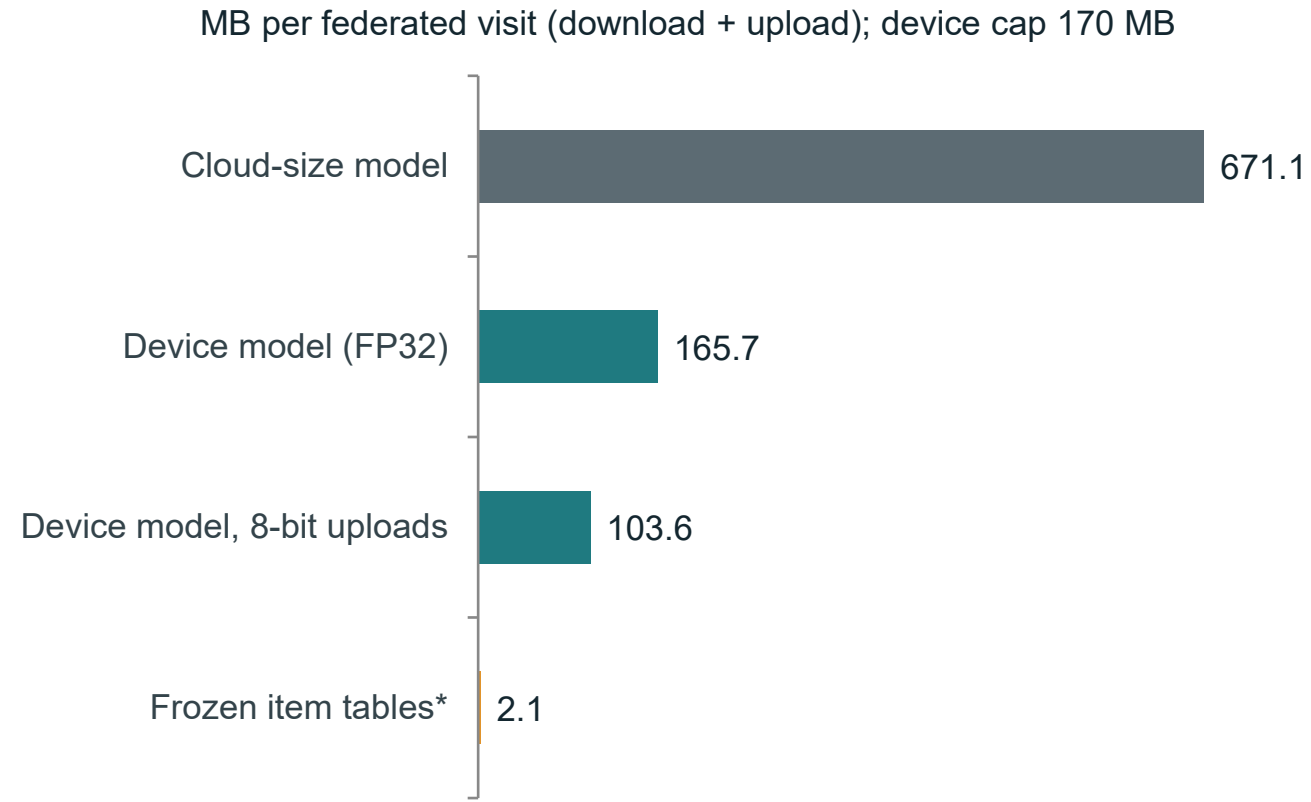
parameters (hidden 256, 3 blocks)

DEVICE MODEL

20.7 M

parameters (hidden 64, 2 blocks); 98% of them are the two item tables

Device download: 82.8 MB (FP32), about 21 MB after INT8 quantisation (estimate until measured).



** 265,162 uploaded parameters ≈ 1.06 MB each way, plus a one-time 81.78 MB table download. To be re-measured.*

Source: Architecture lock (parameters, bytes per visit); realism audit B12, C1; multi-expert review §3.2

WHAT IS COMPARED

Every method, placed by what leaves the device

Privacy level	Methods	What leaves the device	Formal guarantee
Cloud	Large cloud model · small cloud model · full-pool cloud (1.35M users)	Raw events	None
Cloud-trained, runs on device	Small model shipped (ONNX FP32 / INT8) · nearest items on cached embeddings	Raw events, once, for training	None
Pretrained on opted-in users	Pretrained model · + federated on private users · + frozen item tables · + local fine-tune	Opted-in users: events. Private users: updates only	As federated (DP optional)
Federated	FedAvg (ladder-tuned) · FedProx · personal-vector FL · 8-bit uploads · large-cohort FedAvg	Model updates (reveal viewed items)	None; secure aggregation not simulated
Federated + DP	DP-FedAvg at $\varepsilon = 8$ and $\varepsilon = 1$	Clipped, noised updates	Client-level (ε, $\delta = 5e\text{-}6$) DP
Federated + personalisation	Personal vector stays on device · federated model fine-tuned on device	As federated	As federated
Device only	Local-only model (control) · “last viewed item” rule	Nothing	Data never leaves
Reference baselines	Popularity · session kNN · category/brand rule (uses published aggregates)	—	—

Source: Privacy ladder (pre-registration) as revised after the multi-expert review; study description “What is compared”

DISCUSSION / DECISION 2

Which deployment should be the headline answer for Edge?

1

Which row should carry the Edge headline?

Candidates: plain federated · pretrained on opted-in users then federated · cloud model fine-tuned on the device. The independent review recommends the pretrained branch.

2

Should “last viewed item” be the floor every method must beat?

It needs no training and sends nothing. We now report every method's gain over it, with a confidence interval.

3

Is the privacy matrix the right framing for a product audience?

Or would a single ordered ladder (cloud → federated → DP → device) be clearer?

Context: rows as on the previous slide

FEDERATED LEARNING — THE PROBLEM WE FOUND

Untuned federated learning was far below the cloud

≈ 25%

Federated (best tuned stage so far) vs a matched central model on calibration users

0.0288 vs ≈ 0.11–0.12 (central value is an estimate)

≈ 44%

“Last viewed item” rule vs the cloud model on the main validation users

0.0711 vs 0.1612 — no training, nothing uploaded

Different user sets and budgets — a warning sign, not a result.

Diagnosis: the recipe, not a bug

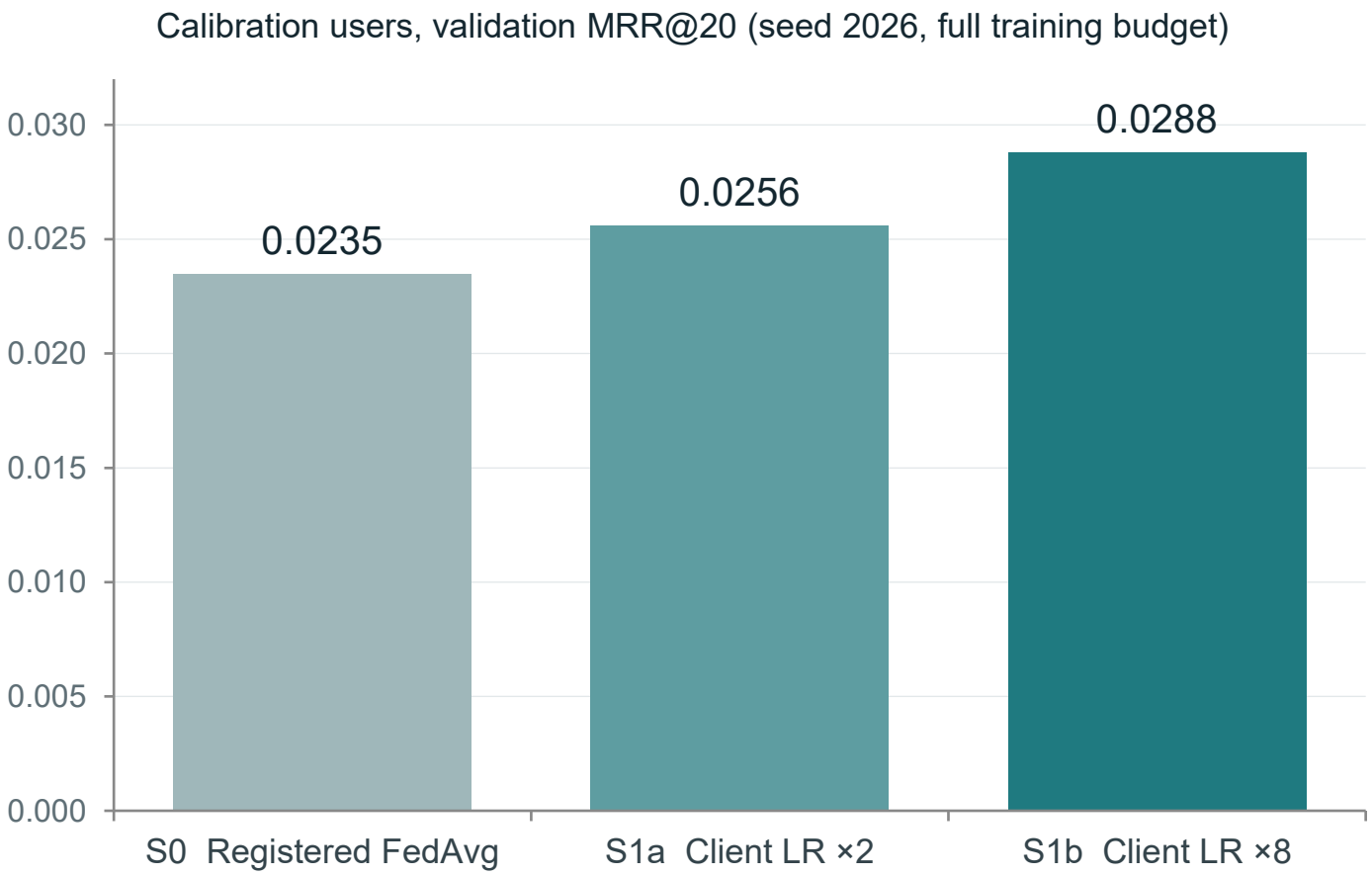
- No server optimizer: plain averaging, server learning rate 1, no momentum.
- Client optimizer restarted every visit: about 4 sign-like steps per visit.
- The cloud learning-rate schedule mapped onto rounds: about half of the budget at ≤ 10% of peak rate.
- Client learning rate too small: the first screen's winner sat at the edge of its grid.
- Both 158k-row item tables start from random, with about 3 visits per user.

Response: a pre-registered improvement ladder — each stage changes one thing, is screened on calibration users only, and is kept only if it gains ≥ 0.002. The untuned start stays in the thesis as the registered reference.

Source: Improvement-ladder register; multi-expert review §1; realism audit §0; ladder research §1.2; contest + baseline values (decision log, 28 Sep)

FEDERATED IMPROVEMENT LADDER

Stages done so far: +22.7% over the registered start



+8.9%

S1a vs S0 — client learning rate grid {0.5, 1, 2}; winner ×2 at the edge → grid extended

+12.7%

S1b vs S1a — extended grid {4, 8, 16}; ×8 wins inside the grid (×16 falls)

+22.7%

S1b vs the registered start

Next stages are shown on the following slide. Main-cohort values of ladder rows are sealed until the final analysis.

Source: Improvement-ladder register (registered calibration screen values); learning-rate selection files

FEDERATED IMPROVEMENT LADDER

Next stages: strictly one at a time, with a stop rule

S2	**Server optimizer**	FedAdam and FedAvg with momentum (incl. small client-LR multipliers)	**Running — 7 screens; ≈ 30 Sep (estimate)**
S3	**LR shape × client optimizer**	Constant rate over rounds; client SGD-momentum vs Adam	**Queued after S2**
S4	**Local computation**	Local passes 1 / 4, batch 8 — one at a time, same exposure	**Queued after S3**
S5	**Longer budget**	Twice the training budget (descriptive budget test)	**After tuning**
S6	**Personalisation**	Personal vector; federated model fine-tuned on device	**Registered arms**
S7	**Start from a pretrained model**	Pretrained on 1.22M opted-in users, then federated	**Pretraining running**
S8	**Freeze the item tables**	Only the encoder is trained and uploaded (≈ 1 MB each way)	**After S7**

Rules

- Each stage builds on the previous winner; no going back.
- A challenger must gain ≥ 0.002.
- Ties within 0.0005 → the simpler option.
- No stage beyond 170 MB per visit or with stateful clients.

Source: Ladder register; ladder amendments (stages S2–S4, S7–S8); decision log 29 Sep (S2 launch 14:44Z, extra runs 17:56Z)

DISCUSSION / DECISION 3

How much effort should federated learning get?

1

Is the staged ladder with a 0.002 stop rule a fair amount of effort for FL?
Or should we stop after the server-optimizer stage and spend the time on the deployment branch?

2

If tuned FL still falls below “last viewed item”, how should we present it?
Our plan: report it plainly as a finding about this recipe at this budget, with the ladder showing what was tried.

3

Is 165.7 MB per visit acceptable for a desktop browser?
Or should the 8-bit and frozen-table variants be the main federated rows?

Context: ladder stages S0–S8; stop rule gain ≥ 0.002

PRIVACY

What the guarantee covers — and what it does not

ε = 8

noise multiplier 0.5642

ε = 1

noise multiplier 1.3094

ε = 4

optional, if time allows

- Client-level differential privacy, $\delta = 5 \times 10^{-6}$: adding or removing one user's whole history barely changes the released model.
- Central DP with a trusted aggregator; Poisson client sampling; Rényi accounting (independently reproduced).
- Not accounted: the federated tuning on calibration users, and the public starting point (catalogue, pretrained weights).

Threat model

Adversary = an honest-but-curious browser vendor (the server). The retailer's own logs are out of scope.

What updates can reveal

- Without DP, an update shows which products the user viewed (the item-table rows that moved).
- Secure aggregation is not simulated — stated as a limitation.
- Frozen item tables remove most, not all, of this leak.

Source: Privacy statement and threat model (registered after the review); independent correctness review of the DP accountant

REALISTIC DEPLOYMENT BRANCH

How a browser would actually ship it

1 Pretrain on opted-in users

1,217,610 users who share data (24.1M decisions); none of them is in the main cohort

2 Federated on private users

Start from the pretrained model; private users send only updates

3 Frozen item tables

Only the small encoder moves: 265,162 parameters ≈ 1.06 MB each way; tables downloaded once

How it is judged (pre-registered, secondary confirmatory)

- Against the full-pool cloud (same opted-in + main users): retention of the pretrained model alone, and of pretrained + federated.
- Value of federated training on private users = (pretrained + federated) − (pretrained alone); frozen vs unfrozen tables with the bytes ratio.
- Status: pretraining running since 29 Sep 13:03 UTC; results sealed until the first validation read.

Source: Warm-start amendment and its review; pretraining cohort build (decision log 29 Sep); multi-expert review §3.2

DISCUSSION / DECISION 4

Privacy level and the deployment story

1

Is DP at ε = 8 and ε = 1 enough, or do we need a curve?

Adding ε = 4 needs one more registered run (~half a day, estimate). A full curve would need several.

2

Is “honest-but-curious browser vendor” the right adversary for Edge?

Should secure aggregation be simulated, or is stating it as a limitation acceptable?

3

Is pretraining on opted-in users a realistic assumption for Edge?

If yes, we would lead the Edge answer with the pretrained → federated → frozen-tables branch.

Context: ε = 8 and ε = 1 registered; ε = 4 optional

ON-DEVICE

What runs entirely on the user's device

Local-only model

Trained from scratch on the user's own 5–100 events. A control, not a product.

Cloud model, fine-tuned on device

Download once, adapt locally; nothing uploaded after the download.

Federated model, fine-tuned on device

Adds local adaptation on top of the federated model.

Pretrained model, fine-tuned on device

The cleanest “download-only” option: its base never saw private users.

Device realism checks

Export	Device model exported to ONNX, FP32 and INT8
Size	82.8 MB FP32; ≈ 21 MB INT8 (estimate)
Latency	Single-thread CPU latency and memory — running
Browser	ONNX Runtime Web in Edge on the owner's laptop — pending

Fine-tuning settings are chosen on a separate panel of unseen users, never on the reported users. Device runs started 29 Sep 16:20 UTC.

Source: Study description “What is compared”; privacy matrix revision; realism audit C1; decision log 29 Sep (device lanes)

COLD START AND USER SEGMENTS

Who benefits, and how much history is needed

12,862

unseen users (never participated) — does the model serve non-participants?

381,295

truly cold users (no training events at all)

16,384 × 2

sparse-history and heavy-history panels

Analyses

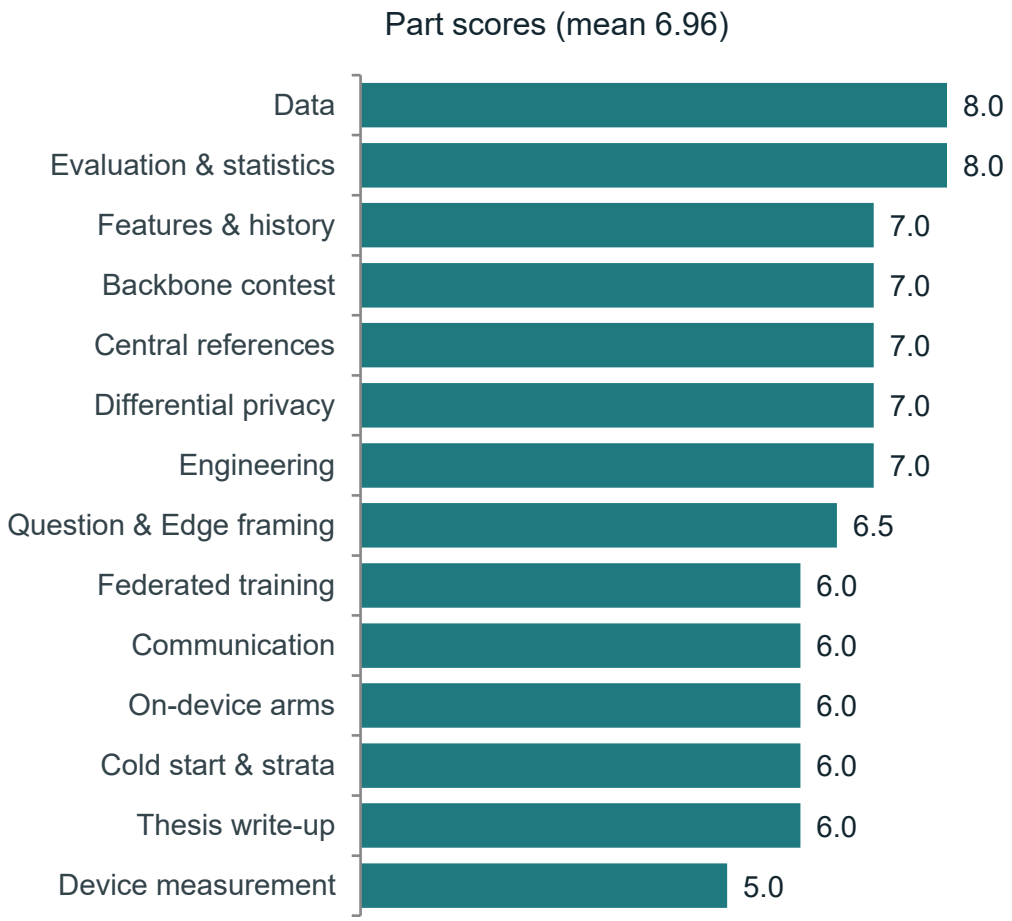
- Quality by history length: 0, 5–10, 11–20, 21–50, 51–100 training events.
- Personas: 3 × 3 grid of category diversity × brand loyalty; activity quartiles.
- Controlled truncation study: cut each user's history to 0, 1, 5, 10, 20, 50 events, with a negative control that must show no change.
- “Events to parity” is claimed only if parity holds at the next history step too.

Honest limit

Grouping by history length is observational: users with long histories differ in other ways. The truncation study is the controlled answer.

Source: Realism audit D2; project map §3; cold-start registration (after the review)

14-part expert review: 7/10 now, 8–8.5 after the fixes



Strengths

- Pre-registration and a single gated test look
- Exact full-catalogue evaluation; no bug found
- Leak-free chronological data protocol
- DP accounting independently reproduced

Weaknesses

- Federated recipe too weak (pre-2019 practice)
- Privacy claims over-stated; no threat model
- Device evidence from a server CPU

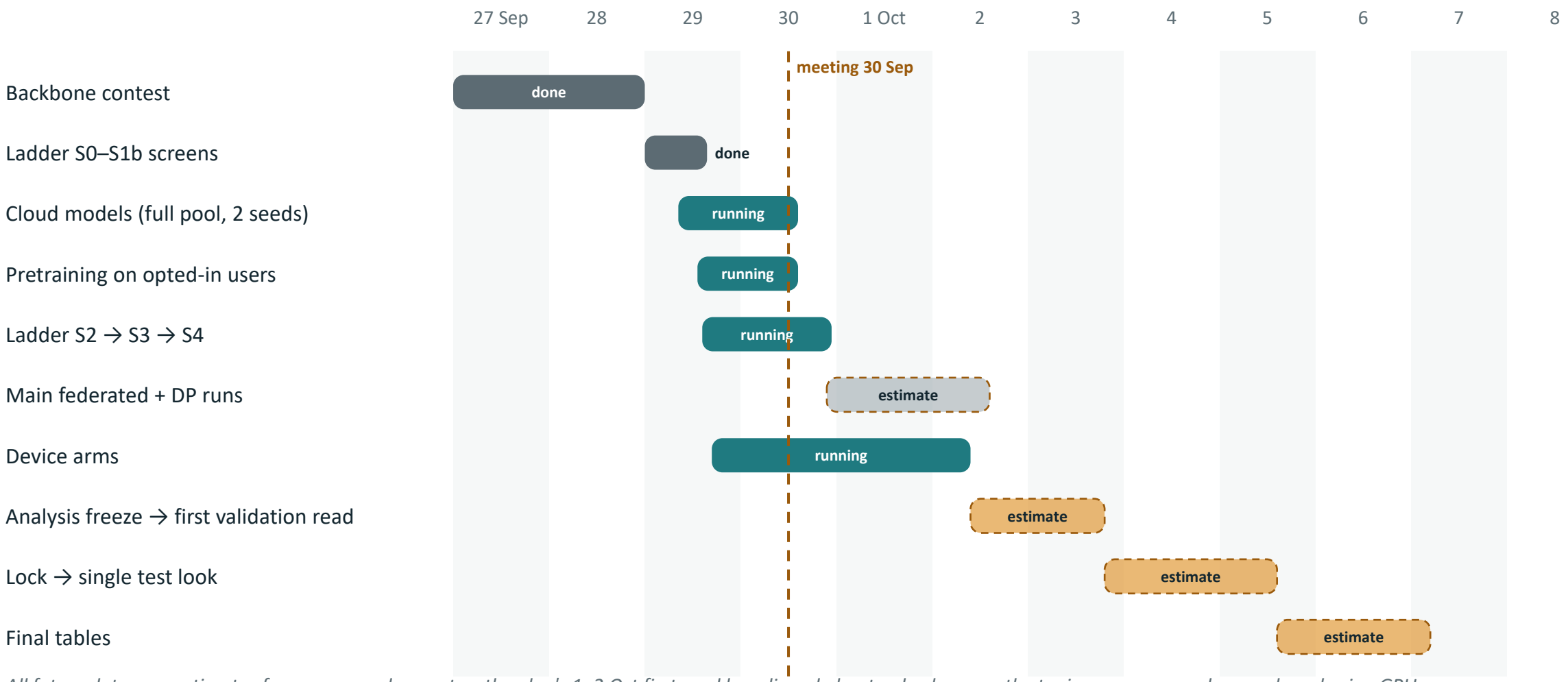
Fixed or registered

- Ladder extended; server optimizers added
- Privacy matrix, threat model, “last item” anchor
- Deployment branch promoted; code blockers fixed

Source: Multi-expert review synthesis (§1 scorecard, §4 gates, §5 strengths); decision log 29 Sep 17:59–18:36Z

TIMELINE

What is running now and when results are expected



All future dates are estimates from measured run rates; the plan's 1–2 Oct first read has slipped about a day because the tuning screens run slower when sharing GPUs.

Source: Master plan v5 (addendum B cost model); ladder research §6 schedule; decision log 29 Sep run status

DISCUSSION / DECISION 5

Priorities if time is short

1

If we must cut, what goes first?
Our pre-registered cut order starts with: extra seeds on panels, the 8-bit second seed, ε = 1, sparse/heavy panels. Never cut: federated ×2, local-only ×2, cloud fine-tune ×2, INT8, the lock, the test look.

2

Is a server-CPU benchmark plus one browser run enough device evidence?
Or should we measure on a second machine or a phone-class device?

3

For cold start, which answer matters more to you?
Observational (by history length) for all methods, or the controlled truncation study for a few methods.

4

Can you pre-approve the Queen's AI-use disclosure, and accept an emailed hash manifest as an external timestamp?
Both are front-matter / integrity items the review asked for.

Context: cut order is pre-registered in the master plan §13

RISKS

Main risks and how each is handled

Risk	Mitigation
Federated looks like a strawman	Pre-registered tuning ladder; every stage recorded; “last item” floor reported
“Federated is not private”	Threat model, privacy matrix, DP arms, frozen-table variant; secure aggregation stated as not simulated
Retention against which cloud?	Both matched and full-pool references reported, the full-pool recipe's upward bias disclosed
Many protocol changes	Every change is a dated, hash-sealed amendment registered before the read it affects
Device claims from a server CPU	Labelled a proxy; browser run on a real laptop; INT8 settings registered exactly
Test week: Black Friday, seen in an earlier phase	Disclosed; Black-Friday sub-period reported; one look only
Time	Pre-registered cut order; anything cut is listed as “registered, not run”

Source: Multi-expert review §6 (defence risks); thesis research review §F; master plan §13

SUMMARY

Where the project stands

DONE

- Frozen data protocol; full-catalogue evaluator
- Backbone contest: SASRec by the registered rule (caveat disclosed)
- Federated ladder S0 → S1b: +22.7%
- Independent reviews; fixes registered before any read

RUNNING

- Full-pool cloud models (2 seeds)
- Pretraining on opted-in users
- Ladder stage S2 (server optimizers)
- Device arms (local-only, fine-tuning)

NEXT

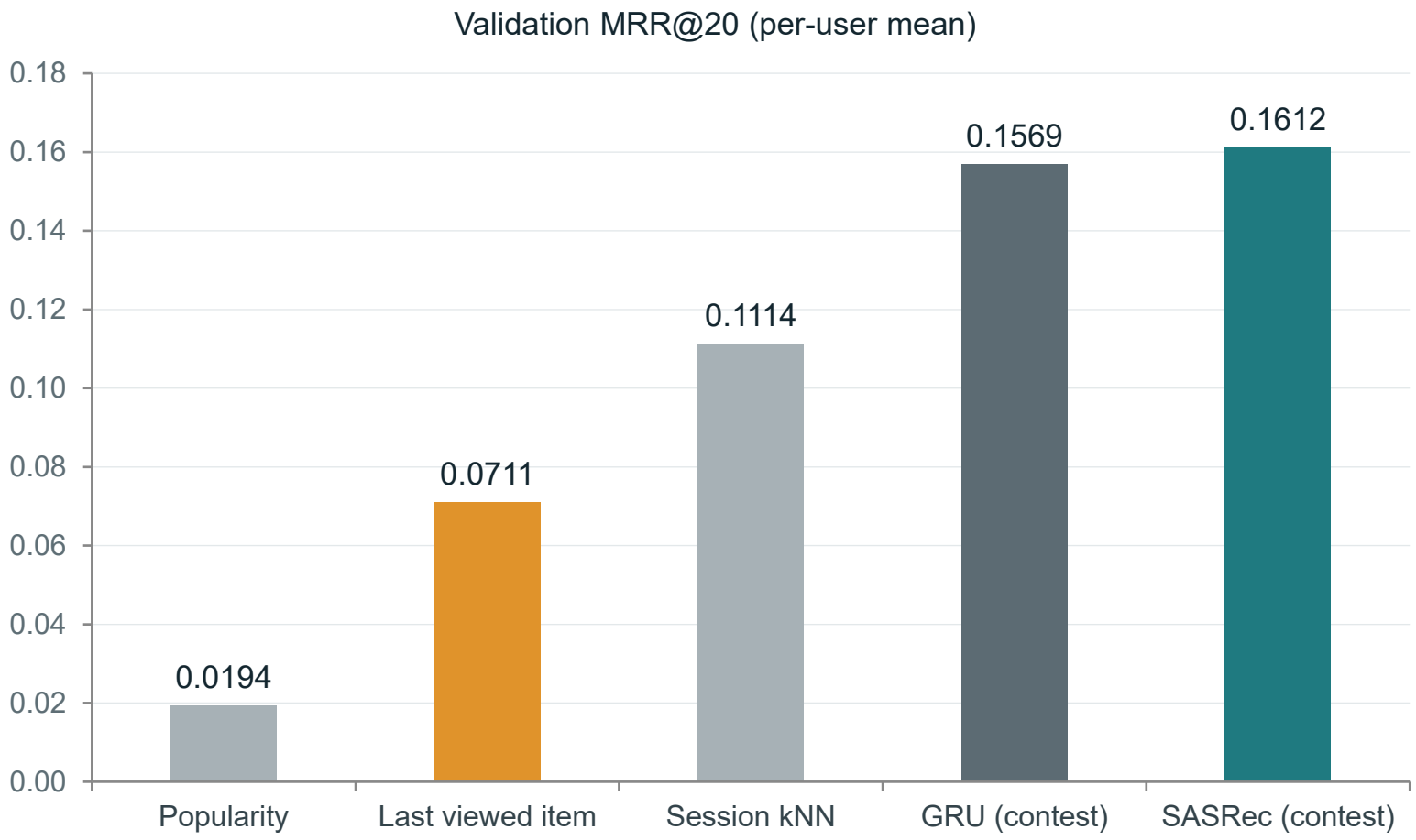
- S3 → S4 → main federated + DP runs
- Analysis freeze → one validation read (≈ 2–3 Oct, est.)
- Lock → single test look (≈ 4–5 Oct, est.)
- Browser benchmark; thesis tables (≈ 6–7 Oct, est.)

Decisions requested today: headline deployment · DP levels · FL effort · cut priorities · AI-use disclosure

APPENDIX

Backup slides for likely questions

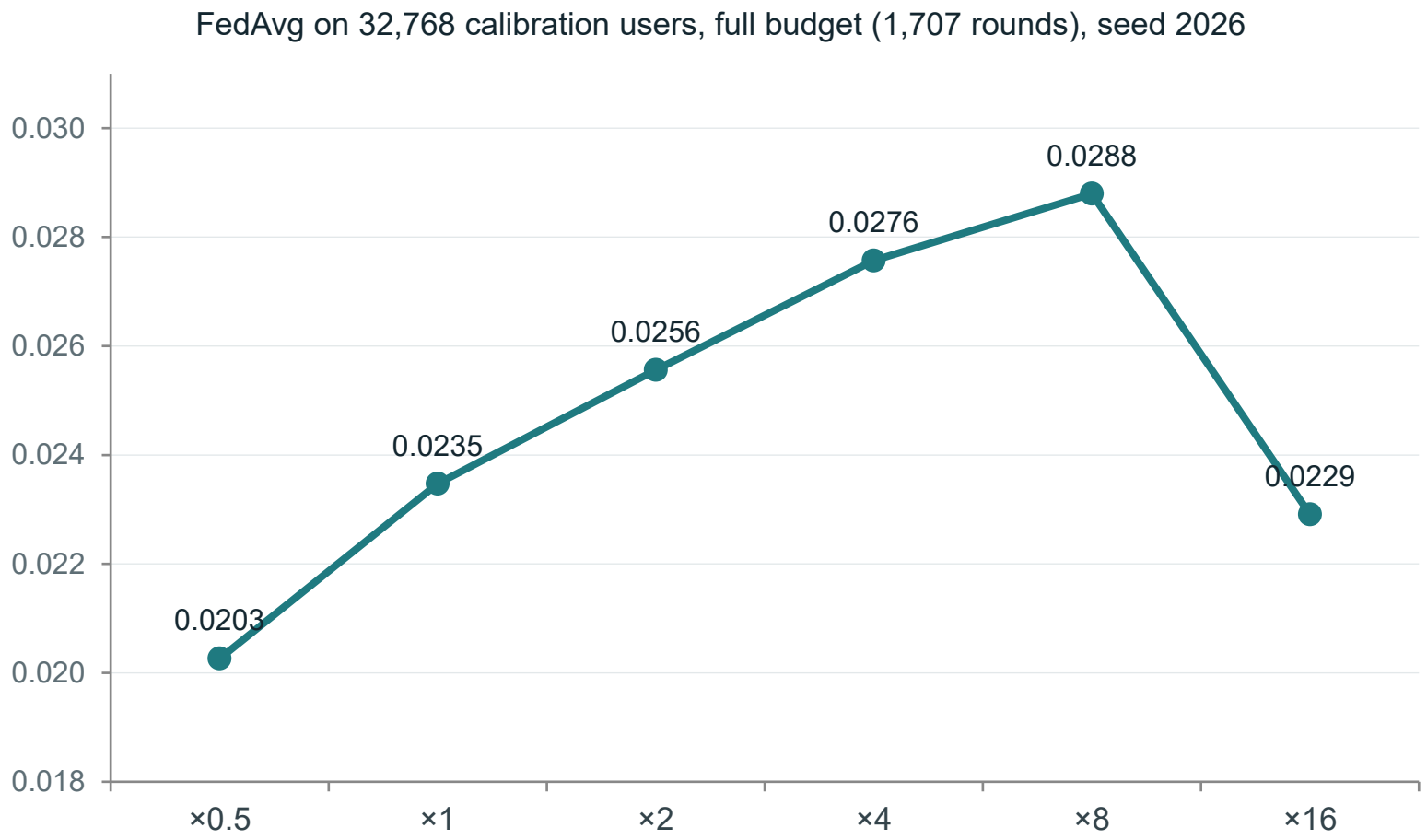
Simple baselines vs the contest models (main validation users)



- NDCG@10: popularity 0.0249, last item 0.1016, session kNN 0.1485.
- “Last viewed item” needs no training and sends nothing — it is the device floor.
- Session kNN reaches about 69% of the contest winner.

Source: Decision log (post-contest reported read of baselines, 28 Sep 11:39Z)

The full screen: rise, peak at ×8, fall



- Multiplier of the client learning rate relative to the small model's central peak rate.
- First screen {0.5, 1, 2} ended at its edge → extension {4, 8, 16} registered before reading.
- ×8 wins inside the grid, so ×32 was not needed.

Source: Learning-rate selection files; improvement-ladder register

Validation MRR@20 per run

Model	Seed	Best checkpoint within budget	Fixed endpoint
GRU, hidden 128	2026	0.15714	0.15714
GRU, hidden 128	2027	0.15673	0.15673
GRU, hidden 256	2026	0.15412	0.15412
GRU, hidden 256	2027	0.15345	0.15345
SASRec, 256 × 3 blocks	2026	0.16120	0.15287
SASRec, 256 × 3 blocks	2027	0.16122	0.15258
SASRec, 256 × 2 blocks	2026	0.16059	0.15086
SASRec, 256 × 2 blocks	2027	0.16006	0.15011

- Within each family, the size was chosen by the 2-seed endpoint mean; between families by the best-checkpoint rule.
- SASRec's best checkpoint is at half the budget; GRU's at the end, so its two columns are equal.
- Winners: GRU hidden 128 and SASRec 3 blocks; SASRec then won between families.

Source: Decision log (reported read of the 8 runs, 28 Sep 11:39Z; contest run 11:25Z)

Mechanism and accounting

Mechanism	DP-FedAvg: per-client clipping of the update, Gaussian noise on the sum
Adjacency	Client-level add/remove (one user's whole history)
δ	5×10^{-6}
Sampling	Poisson, about 1,024 clients per round, 384 rounds; a non-noisy large-cohort twin is run for comparison
Noise multipliers	ε = 8 → 0.5642 · ε = 1 → 1.3094 · ε = 4 → 0.7244 (independently re-derived)
Accounting	Rényi DP; a without-replacement bound reported as sensitivity
Clip norm	Calibrated on a separate run before any DP run; recorded before launch
Not covered	Tuning on calibration users; public bootstrap (catalogue, pretrained weights)

Source: Independent federated-code review (DP accountant check); privacy statement registration; project map (DP settings)

APPENDIX E — TERMS

Plain-language glossary

Retention	Quality of a method ÷ quality of the cloud reference, × 100
MRR@20	Mean reciprocal rank of the true next product within the top 20, averaged per user
Calibration users	32,768 users used only for tuning screens; their values may be read before the final analysis
Pre-registered	Rule written and hash-sealed before the result it governs is seen
Sealed test week	23–29 Nov 2019; opened once, after all methods are locked
Training budget	Six passes' worth of exposure over the cohort's data, equal for all methods
Ladder stage	One registered change to the federated recipe, kept only if it gains ≥ 0.002
Frozen item tables	Product tables fixed after pretraining; only the small encoder is trained and uploaded

Source: Deck data file: company/presentations/SUPERVISOR_DECK_DATA.md (every number with its source)